LegalBot — Docker Containerization Lab Manual

GenAI & ML DevOps Lab — Containerization Experiment

Supported OS: Windows (PowerShell/CMD) | macOS (Apple Silicon/Intel) | Linux (Ubuntu/Debian)

1. Prerequisites & Verification

Ensure Docker Desktop (Windows/macOS) or Docker Engine (Linux) is installed and running. Test via terminal:

```
docker --version
docker compose version
```

2. Step 1: Create .dockerignore

Create a file named .dockerignore in the project root (LegalBot/) to exclude virtual environments, cache, and secrets:

```
.venv/
venv/
env/
__pycache__/
*.py[cod]
.pytest_cache/
.git/
.gitignore
.vscode/
.idea/
.env
*.md
DOCKER_LAB_GUIDE.*
.DS_Store
Thumbs.db
```

3. Step 2: Create Dockerfile

Create a file named Dockerfile (no file extension) in the project root (LegalBot/):

```
FROM python:3.11-slim

ENV PYTHONDONTWRITEBYTECODE=1 \
    PYTHONUNBUFFERED=1 \
    PYTHONPATH=/app

WORKDIR /app

RUN apt-get update && \
    apt-get install -y --no-install-recommends curl && \
```

```
    rm -rf /var/lib/apt/lists/*

COPY requirements.txt /app/requirements.txt
RUN pip install --no-cache-dir --upgrade pip && \
    pip install --no-cache-dir -r requirements.txt

COPY app/ /app/app/
COPY streamlit_app.py /app/streamlit_app.py

EXPOSE 8000 8501

HEALTHCHECK --interval=30s --timeout=5s --start-period=10s --retries=3 \
    CMD curl -f http://127.0.0.1:8000/health || exit 1

CMD ["python", "-m", "uvicorn", "app.main:app", "--host", "0.0.0.0", "--port", "8000"]
```

4. Step 3: Build Docker Image

Run from the LegalBot/ directory containing the Dockerfile:

```
docker build -t legalbot:latest .
docker images legalbot
```

Note for macOS (Apple Silicon): Add '--platform linux/amd64' if cross-compiling for x86.

5. Step 4: Run the Container

Pass your API keys dynamically using --env-file .env (never bake keys into the image):

Mode A: FastAPI Backend & Web UI (Port 8000 — Default)

```
docker run -d --name legalbot-app -p 8000:8000 --env-file .env legalbot:latest
```

Mode B: Streamlit Dashboard (Port 8501 — Alternative)

```
docker run -d --name legalbot-streamlit -p 8501:8501 --env-file .env legalbot:latest streamlit run
streamlit_app.py --server.port=8501 --server.address=0.0.0.0
```

6. Step 5: Verify & Test

1. Check running status: docker ps

2. View live logs: docker logs -f legalbot-app

3. Open browser URLs:

- FastAPI Web UI: http://localhost:8000

- Interactive Swagger Docs: http://localhost:8000/docs
- Health Check: http://localhost:8000/health
- Streamlit UI (Mode B): http://localhost:8501

4. Terminal health verification:

```
# Windows (PowerShell):
Invoke-RestMethod -Uri http://localhost:8000/health | ConvertTo-Json

# macOS & Linux:
curl http://localhost:8000/health
```

7. Container Lifecycle Commands

Action	Command
List running containers	`docker ps`
List all containers	`docker ps -a`
Stream live logs	`docker logs -f legalbot-app`
Stop container	`docker stop legalbot-app`
Start stopped container	`docker start legalbot-app`
Restart container	`docker restart legalbot-app`
Execute shell inside container	`docker exec -it legalbot-app /bin/bash`
Remove container	`docker rm -f legalbot-app`
Remove image	`docker rmi legalbot:latest`

8. Quick Troubleshooting

Issue	Cause	Solution
Port 8000 in use	Local uvicorn process running on host.	Stop host uvicorn, or run with -p 8080:8000.
Docker daemon closed	Docker Desktop application not started.	Start Docker Desktop; wait for engine running icon.
API key not configured (503)	Missing --env-file or empty keys.	Verify .env contains valid keys and include --env-file .env.

Windows line endings	.env has Windows CRLF line endings.	Open .env in VS Code and change line endings to LF.
Linux permission denied	Non-root user not in docker group.	Run: sudo usermod -aG docker $USER && newgrp docker.